

Electrical Science

ENGR 2303

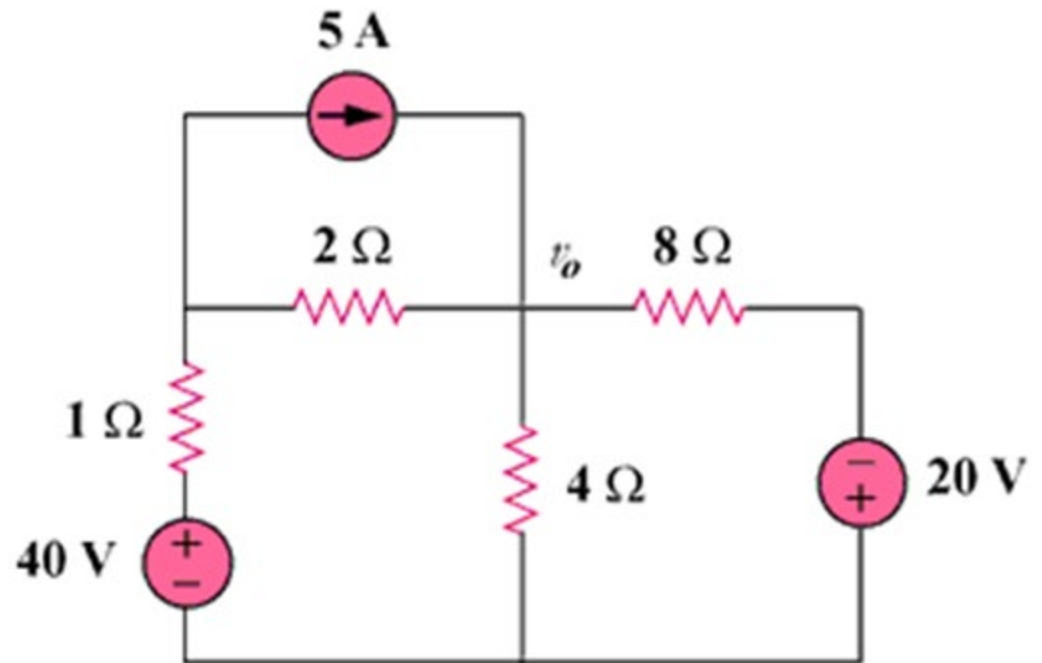
Mesh Analysis

Video 5

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Mesh Analysis

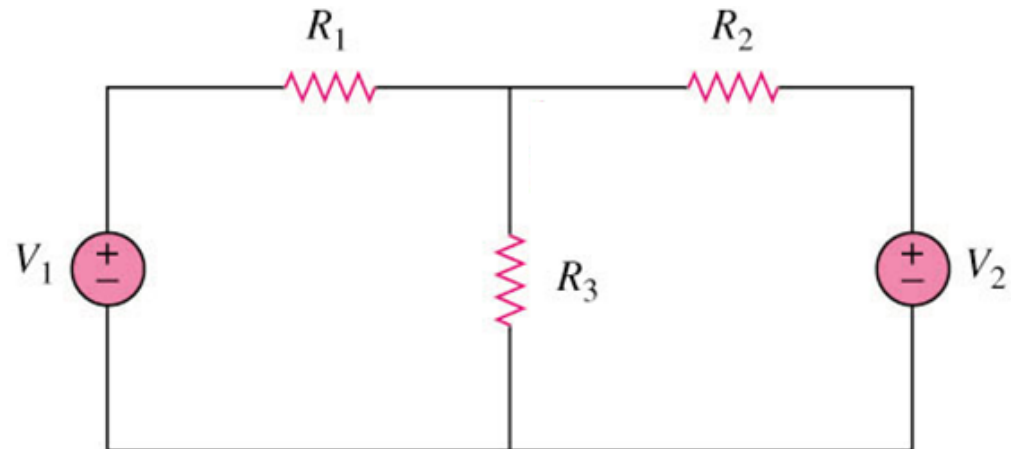
- Mesh analysis.
 - a loop which does not contain any other loops within it.
- Mesh analysis with current source..
- Examples



Mesh Analysis

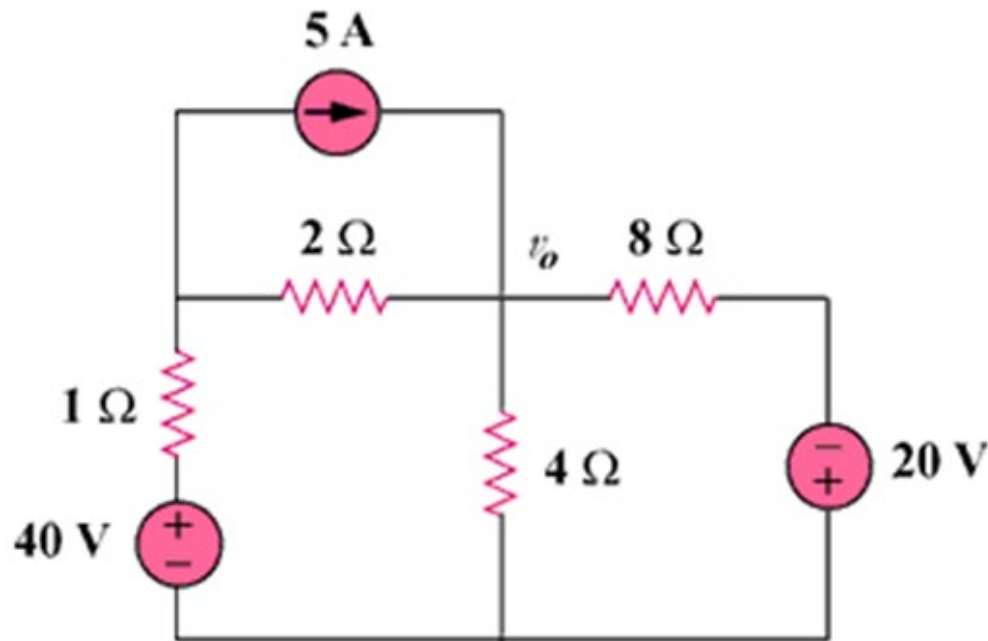
Steps to determine the mesh currents:

1. Assign mesh currents $i_1, i_2, \dots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.



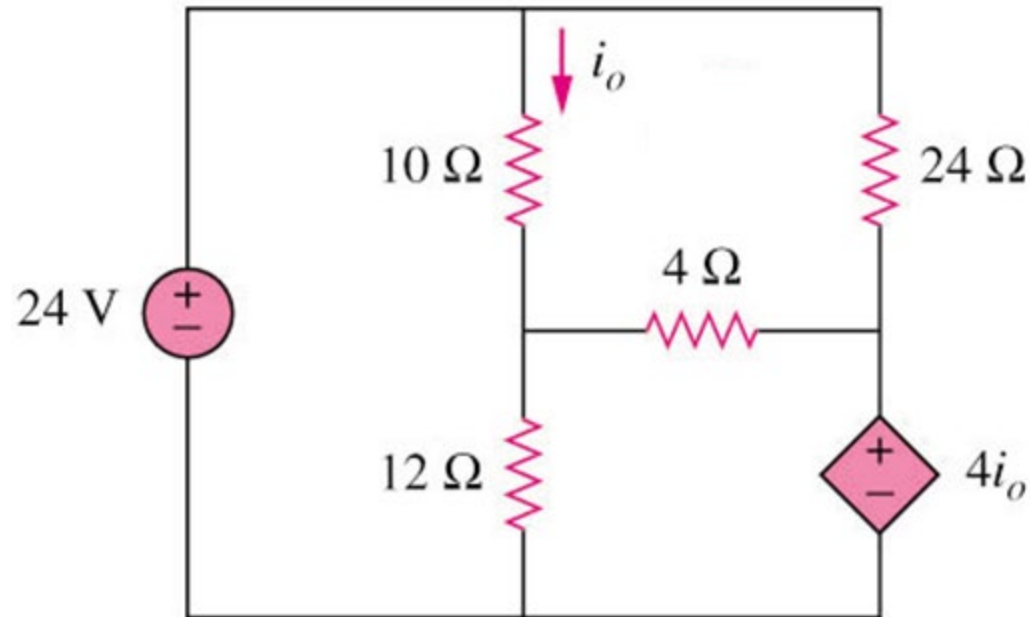
Mesh Analysis with Independent Sources

Apply mesh analysis to find v_o in the circuit

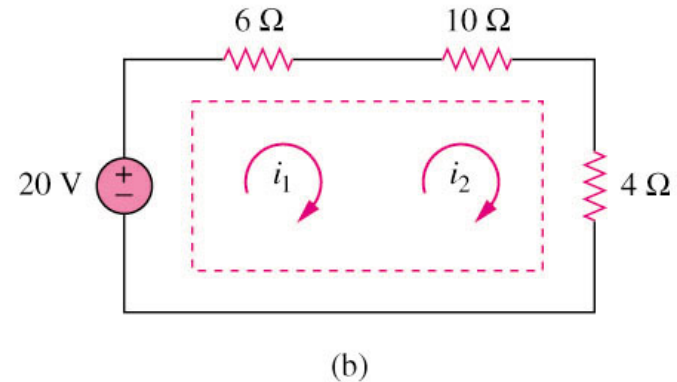
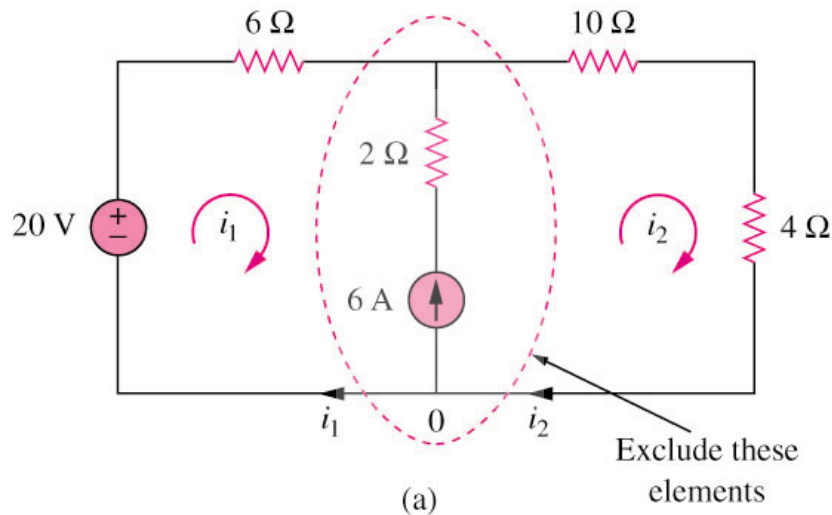


Mesh Analysis with Dependent Source

Example: Find the current I_o in every resistors.



Mesh Analysis with Current Source



A **super-mesh** results when two meshes have a (dependent or independent) current source in common as shown in (a).

1- We create a super-mesh by excluding the current source and any elements connected in series with it as shown in (b).

2- Write KVL in the super-mesh using the loop currents of the two adjacent meshes.

3- The sum or the difference of the two mesh currents equal the current source in the common branch between the two meshes.

Mesh Analysis with Current Source

Example: Find the power delivered by the voltage and current sources.

